

1.

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

1. **Motherboard** – The motherboard is the main circuit board of the computer. It connects all the components and allows them to communicate with each other.
2. **CPU (Processor)** – The CPU is the brain of the computer. It processes instructions and performs all calculations needed to run programs.
3. **RAM (Random Access Memory)** – RAM is temporary memory that stores data while programs are running. More RAM helps the computer work faster and handle multiple tasks.
4. **Storage (SSD/HDD)** – The SSD or HDD stores the operating system, software, games, and personal files. An SSD is faster than an HDD.
5. **Power Supply (SMPS/PSU)** – The power supply converts electricity from the wall into the correct power needed by all the computer components.
6. **GPU (Graphics Card)** – The GPU processes graphics and displays images on the monitor. It is important for gaming, video editing, and other graphics-related work.

7. **CPU Cooler (Fan & Heatsink)** – The CPU cooler removes heat from the processor. It keeps the CPU at a safe temperature and prevents overheating.
8. **Case Fans** – Case fans move cool air into the cabinet and push hot air out. This helps keep all the internal parts cool.
9. **Power and Data Cables** – These cables connect the components inside the PC. They carry power from the power supply and transfer data between devices.

2.

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



1. Unplug the Power Cable
First, turn off the PC and unplug the power cable from the back of the power supply.



2. Remove the Side Panel
Remove the screws at the back of the cabinet and slide the side panel off.



3. Disconnect All Cables
Carefully disconnect all power and data cables connected to the motherboard, storage, and other components.



4. Remove RAM
Press the clips on both sides of the RAM stick and gently pull it out from the slot.



5. Remove HDD/SSD
Disconnect the data and power cables from the drive. Remove the screws or release the bracket and take out the drive.



6. Remove GPU (Graphics Card)
Unscrew the bracket, disconnect the power connector (if any) and gently pull out the graphics card from the PCIe slot.



7. Remove CPU Cooler
Disconnect the cooler cable from the motherboard and remove the screws or clips. Then lift the cooler off the CPU.



8. Remove Motherboard
Remove all screws holding the motherboard. Carefully lift it up and take it out of the cabinet.

3.

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

➤ **Step 1: Install the Motherboard**

I placed the motherboard inside the cabinet and tightened the screws. This is done first because all other components connect to the motherboard.

➤ **Step 2: Install the CPU Cooler**

I attached the CPU cooler to the processor and connected its fan cable to the motherboard. This keeps the CPU cool while the computer is running.

➤ **Step 3: Install the RAM**

I inserted the RAM into the memory slots until the clips locked. RAM is needed before the system can start properly.

➤ **Step 4: Install the HDD/SSD**

I placed the HDD/SSD in its bay and connected the SATA data cable and power cable. This allows the computer to store and access data.

➤ **Step 5: Install the Graphics Card (GPU)**

I inserted the GPU into the PCIe slot and secured it with screws. This is needed for displaying graphics and running games.

➤ **Step 6: Connect All Power and Data Cables**

I connected the motherboard power cable, CPU power cable, SATA cables, and front panel cables. These connections allow all components to receive power and communicate.

➤ **Step 7: Close the Side Panel**

I put the side panel back on and tightened the screws. This protects the internal components from dust and damage.

➤ **Step 8: Power On and Test**

I connected the power cable and turned on the PC. The computer started successfully, confirming that all components were installed correctly.

4.

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

➤ **Safety Precautions**

☐ **Turn off the PC and unplug the power cable**

Why: Prevents electric shock and protects the components.

☐ **Press the power button for a few seconds after unplugging**

Why: Removes any remaining electricity inside the PC.

☐ **Work on a clean and flat surface**

Why: Keeps the components safe and prevents them from falling.

☐ **Avoid touching the gold connectors or chips**

Why: Helps prevent damage from static electricity or fingerprints.

☐ **Keep screws in a small container**

Why: Prevents screws from getting lost during the process.

➤ **Tools Needed**

☐ **Screwdriver**

Why: Used to remove and tighten screws inside the PC.

☐ **Anti-Static Wrist Strap (Optional)**

Why: Protects sensitive computer parts from static electricity.

☐ **Small Container for Screws**

Why: Keeps all screws organized and easy to find.

☐ **Soft Brush or Compressed Air**

Why: Removes dust from fans and other components safely.

☐ **Flashlight (Optional)**

Why: Helps you see small connectors and screws inside the cabinet.